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SUPERSEDING
MIL-R-12544 (SigC)
18 March 1953

MILITARY SPECIFICATION

CHARGER, BATTERY, METALLIC (RECTIFIER RA-63-( ))

This specification has been approved by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 Scope.—This specification covers one type of portable, selenium plate, full-wave bridge rectifier designated as Rectifier RA-63-( ). (See 6.3.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 The following specifications, standards, and drawings of the issues in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

NN-B-591—Boxes; Fiberboard, Wood-Cleated (For Domestic Shipment).
NN-B-601—Boxes; Wood-Cleated-Plywood, For Domestic Shipment.
NN-B-621—Boxes; Wood, Nailed and Lock-Corner.
LLL-B-636—Boxes; Fiber, Corrugated (For Domestic Shipment).
UU-T-111—Tape, Paper, Gummed (Sealing and Securing).

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JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood Cleated, Plywood (For Boxes Whose Weight of Contents Does Not Exceed 1,000 Pounds).
JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood Nailed (For Weight of Contents Not in Excess of 1,000 Pounds).
JAN-P-108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116—Preservation, Methods of.
JAN-P-117—Packaging and Packing for Overseas Shipment—Bags, Interior Packaging.
JAN-P-127—Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL-B-131—Barrier-Material; Water-Vaporproof, Flexible.
JAN-T-152—Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment, General Process for.
MIL-V-173—Varnish, Moisture- and Fungus-Resistant, for The Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-I-631—Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
JAN-T-713—Twine, Lacing and Typing, Electrical and Electronic Equipment.
MIL-V-6894—Varnish, Spar, Glyceryl-Phthalate.
MIL-R-10187—Reference Symbols for Signal Corps Equipment.
MIL-T-10513—Tropicalization of Materials Used in Signal Corps Equipment.
MIL-R-11050—Rectifiers, Selenium (Open Construction).
MIL-E-11857—Enamel, Synthetic, Semi-Gloss.
MIL-I-16557—Ink, Marking, Quick-Drying (For Non-Porous Surfaces).

U. S. ARMY

3-174—Enamel, Synthetic, Semi-Gloss.

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19–10—Filler, Interior Packing, Fiberboard, Corrugated, Single and Double Faced, For General Packing Purposes.
71–4944—Identification Marking of Signal Equipment.

STANDARDS

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking of Shipments.
MIL–STD–130—Identification Marking for U. S. Military Property.

DRAWINGS

SIGNAL CORPS

460–12—Group B–2, Sampling Plans.
SC–D–15914 — Moisture-Resistance Test Cycle for Ground Signal Equipment.
SC–C–20123—Rectifier RA–63–( ), Drawing and Data List.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting offier. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Preproduction samples.

3.1.1 *Quantities.*—After award of contract, two complete Rectifiers RA–63–( ) and three transformers shall be furnished for preproduction testing.

3.1.2 *Compliance with specified requirements.*

3.1.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.1.2.3 for exceptions.)

3.1.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.1.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.1.2.3 When deviation from 3.1.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and the manner in which it will be corrected in production of equipment on contract.

3.1.3 *Test data.*—The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.1.4 *Reference standards.*—After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any difference of opinion regarding interpretation of requirements.

3.2 Plastic material.—Where not machined, plastic material shall have the original smooth or polished surfaces. Surfaces that have been cut, sawed, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the type of material used and the intended application.

3.3 Soldering.—Soldered connections shall be neat. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Wherever practicable, excess rosin shall be removed with a wire brush and then a dry cloth; any resulting loose flakes of rosin shall be carefully removed from the inside of the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.1 *Flux and cleaning agents.*—Flux for soldering shall be rosin or rosin and alcohol. No acid or acid salts shall be used in preparation for or during soldering; however, excep-

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tion is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.4 Wiring and cabling.—Wiring and cabling shall be neat and sturdy.

3.4.1 *Cabling.*—Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is mechanically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance of the equipment. The cabling of wires shall be effected by twine conforming to Specification JAN–T–713.

3.4.2 *Slack.*—Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.4.3 *Protection.*—Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges, under specified service conditions. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal (partitions, shields, and the like), they shall be protected by suitable grommets or bushings.

3.4.4 *Support.*—Wires and cables shall be properly supported and secured to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equivalent has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.4.5 *Clearance.*—Clearance between wires or cables, and parts such as resistors and other heat generating parts, shall be sufficient to avoid deterioration of the wires or cables because of the heat dissipated by such parts under specified service conditions for the equipment. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be at least ¼ inch unless use of specified parts makes such clearance impracticable.

3.4.6 *Splicing.*—Wires in a continuous run between two terminals shall not be spliced during assembly of the equipment, except where a stranded conductor is spliced to a solid conductor and the two are supported at the splice.

3.4.7 *Insulating sleeving.*—Insulating sleeving shall not be used unless specified for a particular application or specifically approved by the contracting officer. Approval will not be granted when other means of insulation are practicable. If insulating sleeving is used, it shall be a suitable class or type conforming to Specification MIL–I–631.

3.4.8 *Connections.*—Soldering leads, lugs, and terminals shall be tinned, silver-plated, or lead-alloy coated. Wires subject to breakage at the connections shall be provided with terminals that grip the wire insulation. Solderless terminals may be used on tinned, stranded wire in power circuits, but other wire connections shall be soldered. Where practicable, wires soldered to terminals shall be looped at least once and not more than twice around the terminals before soldering, or equivalent means shall be employed. Textile insulation ends of wires shall be secured against fraying by mechanical means or by application of varnish or lacquer conforming to Specification MIL–V–173.

3.5 Marking.

3.5.1 *Identification marking.*—Identification marking of components and parts shall be as specified by the applicable documents. When identification marking is not so specified, the following requirements shall apply:

3.5.1.1 *Nomenclature marking.*—Components and parts to which Joint Army-Navy or Signal Corps nomenclature has been assigned shall bear identification marking in accordance with Specification 71–4944. For Air Force purchases, identification marking shall be in accordance with Standard MIL–STD–130.

3.5.1.2 *Other identification marking.*—Identification marking of components and parts except as specified in 3.5.1.1, shall conform to the following:

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(a) *Electrical articles.*—Electrical articles shall be marked with the manufacturer's name or trademark, his identifying part number, value and rating when practicable, and other pertinent information. When space is limited, as much as possible of the foregoing shall be included, preference being in the order given. Subassemblies that are replaceable as a unit shall be marked in the same manner, but the marking shall be such as not to be confused with the marking for any part of the subassembly.

(b) *Nonelectrical articles.*—Nonelectrical components shall be marked with the manufacturer's name or trademark, his identifying part number, and other pertinent information. Parts such as screws, nuts, washers, and the like, need not be marked but larger or more complicated replaceable parts shall be marked, if practicable, as specified for components. Subassemblies that are replaceable as a unit shall be marked in the same manner as parts, but the marking shall be such as not to be confused with the marking for any part of the assembly.

Size of the marking shall provide maximum legibility commensurate with the size of the article and the space available for marking.

3.5.2 *Marking of type numbers and reference designations.*

3.5.2.1 *Reference designations.*—Reference designations, as specified herein, shall conform to Specification MIL–R–10187. Reference designations shall not be marked on spare parts.

3.5.2.2 *Other electrical parts.*—The reference designation for each part having electrical circuits or connections shall be marked on an adjacent surface; but if space is not available there, the reference designation shall be marked on the part itself.

3.5.2.3 *Nonelectrical parts.*—The reference designation of each nonelectrical part except screws, nuts, washers, bushings, and similar small hardware shall be marked on the chassis, frame, panel, etc., immediately adjacent to the part; but if space is not available there, the reference designation shall be marked on the part itself.

3.5.3 *Method of marking.*—Marking shall be applied by direct application or by label. Direct application marking shall be one of the following methods, applied in such manner as to be legible and permanent:

Group I: Die stamping, engraving, etching, molding, steel stamping.

Group II: (for stenciling): Semi-gloss synthetic enamel, Grades I or III, conforming to Specification MIL–E–11857; quick drying marking ink (for nonporous surfaces) conforming to Specification MIL–I–16557.

Group III: Permanent ink, rubber stamping, silk screening.

Group III markings, and pigment-filled Group I markings, shall be protected by a clear varnish conforming to Specification MIL–V–6894, except where markings are coated during overall tropicalization treatment of the equipment. (See 3.7.)

3.6 Cleaning.—Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipments. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.7 Tropicalization.—The assembled equipment shall be treated in accordance with Specification JAN–T–152. Before being assembled into the equipment, materials and parts shall be treated in accordance with Specification MIL–T–10513.

3.8 Service conditions.—The equipment shall meet the following service conditions:

3.8.1 *Equipment operating.*—The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) *Operation.*—Operation from 105 to 125 volts, 50–60 cycle, single-phase a-c and charging 6 fully-discharged cells in

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series (2 batteries BB–55 in series) starting at an initial rate of 15 to 16 amperes.

(b) *Temperature.*—Any ambient temperature in the range of −40 F to +125 F.

(c) *Relative humidity.*—Any relative humidity up to 95 percent.

3.8.2 *Equipment nonoperating.*—The equipment shall meet the requirements of 3.8.1 after subjection to any one or more of the following nonoperating conditions.

(a) *Temperature.*—Continuous exposure for 72 hours in an ambient temperature of −80 F, and 4 hours at +160 F, at any orientation.

(b) *Relative humidity.*—Relative humidity up to 100 percent, including condensation caused by temperature changes.

(c) *Orientation.*—Indefinite storage in any orientation.

(d) *Vibration and bounce.*—See 3.14.

3.9 Construction.—Rectifier RA–63–( ) shall be constructed in accordance with Drawing and Data List SC–C–20123 and the requirements specified herein.

3.9.1 *Rectifier stacks.*—Rectifier stacks shall be in accordance with the requirements of Specification MIL–R–11050.

3.10 Input.—The rectifier shall be capable of operation from an input of 105 to 125 volts, 50 to 60 cycles, single-phase a-c.

3.11 Output.—When operating from inputs as specified above, the rectifier shall be capable of charging 6 fully-discharged lead-acid type storage cells in series (2 batteries BB–55 in series) at the initial rate of approximately 15 amperes with the control switch in the "HI-CHG" (High Charge) position. The rectifier shall also be capable of charging 6 fully-discharged lead-acid cells in series at the rate of approximately 5 amperes with the control switch in the "TR-CHG" (Trickle-Charge) position.

Note. As a reference point, a 6-cell (12-volt) battery is to be construed as being fully-discharged when it has been discharged to an end voltage of 10.5 volts at the 20-hour discharge rate.

3.12 Efficiency.—The operating efficiency of Rectifier RA–63–( ) shall not be less than 60 percent.

$$\frac{\text{OUTPUT-Power}}{\text{INPUT-Power}} = 0.6 \text{ or better}$$

3.13 Bounce conditioning.—After final assembly, each equipment on the contract shall be placed in its normal operating position on the table of the Package Tester as made by the L. A. B. Corporation, Summit, New Jersey, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$ inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this conditioning for 1 minute. After conditioning, the equipment shall be capable of meeting the inspection of Section 4.

3.14 Vibration and bounce.

3.14.1 *Vibration.*—The equipment shall not suffer any mechanical or electrical damage after subjection to the test of 4.4.

3.14.2 *Bounce.*—The equipment shall meet the tests of 4.5.

3.15 Interchangeability.— Corresponding components on contract shall be physically and functionally interchangeable as complete units. Components shall be so constructed that any subassembly or part meeting specified dimensions, ratings, characteristics, etc., can be substituted for a corresponding, replaceable subassembly or part without modification of the substituted unit or of the equipment and without causing the equipment to deviate from specified requirements. When dimensions, ratings, characteristics, etc. are not specified the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are replaceable shall be as determined by the contracting officer.

3.16 Technical literature.—Technical literature shall be furnished as specified by the contract. (See 6.2.)

3.17 Workmanship.—The equipment, subassemblies, and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 Inspection, general.—Inspection shall consist of Group A and Group B inspection as

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specified in 4.1.1 through 4.1.3.1. (See 6.2.) Group B inspection shall be performed on lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. Samples for inspection shall be selected in an unbiased manner by the Government inspector.

4.1.1 *Group A inspection.*—Group A shall consist of Group A inspection as specified on Table I and shall be performed by either of the following methods or a combination of both:

(a) 100 percent inspection.

(b) Statistical sampling in accordance with Standard MIL–STD–105. The acceptable quality levels (AQL's) shall be as specified in Table II or the contract.

4.1.1.1 *Order of inspection within group A.*—Group A inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.2 *Group B inspection.*—Group B shall consist of the Group B inspection specified in Table I.

4.1.2.1 *Sampling procedure.*—Sampling plans for Group B inspection shall be as specified in Table I or in the contract and shall be in accordance with Drawing 460–12 for Group B–2 sampling plans. Unless otherwise specified, inspection shall be started using Level II of Drawing 460–12. If the number of defective sample units from a lot exceeds the acceptance number specified in Drawing 460–12, the lot shall be rejected and the contractor shall take immediate action to correct the cause of failure.

4.1.2.2 *Order of inspection within Group B.*—Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.3 *Disposition of rejected product.*—When rejected product (sample unit, lot, or other portion of product) is submitted for inspection, the product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.1.3.1 *Groups A and B.*—Disposition of product rejected during Group A or Group B inspection shall be in accordance with Standard MIL–STD–105.

4.2 Visual and mechanical inspection.—Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be examined for compliance with specified requirements. Examination also may be made for other visual or mechanical defects similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of Standard MIL–STD–105.

4.3 Moisture-resistance test.

Table I.—Inspection procedures

Inspection	Req. par.	Test par.	Preproduction	Group A AQL	Group B plans
Visual and mechanical, major		4. 2	X	1.5 DPHU*	
Visual and mechanical, minor		4. 2	X	6.5 DPHU*	
Load	3. 11	4. 8	X	1.5 DPHU*	
Operation	3. 11	4. 7	X		B–2
Moisture-resistance	3. 8	4. 3	X		
Vibration	3. 14	4. 4	X		
Bounce	3. 15	4. 5	X		
Extreme temperature	3. 8	4. 6	X		

* Defects per hundred units.

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4.3.1 *Test conditions.*

(a) Do not move components from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.3.2 *Test procedure.* — The components shall be tested as follows:

(a) Dry at 130°±5° for 24 hours.

(b) Condition at 77°±5° F, and 40 to 50 percent relative humidity for 24 hours.

(c) Measure performance and readjust if necessary to meet specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing SC–D–15914. Performance of the equipment shall be measured during the periods shown on the drawings and shall conform to specification requirements.

(e) After cycling has been completed, condition the equipment for 24 hours at 77°±5° F, and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specification requirements.

4.3.3 *Failure.*—If any component fails to meet specification requirements during cycling, or after the test, it does not pass the test.

4.4 **Vibration test.**—The equipment shall be tested for compliance with 3.14 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. The vibration table shall provide approximately sinusoidal vibration.

(b) The equipment shall be vibrated successively for at least 30 minutes in each of three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

4.5 **Bounce test.**—The equipment, as packaged for shipment, shall be placed on the table of the Package Tester as made by the L. A. B. Corporation, Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ±1 percent for a total of 3 hours. During each ½-hour period of the test, the packaged equipment shall rest on a different face.

4.6 **Extreme temperature tests.**

4.6.1 *Low temperature test.*—The rectifier shall be placed in a cold chamber, at an ambient temperature of −80° F, for a period of at least 72 hours. The temperature shall then be raised to −40° F, and the rectifier allowed to reach thermal equilibrium. After exposure to the −40° F ambient temperature for at least 24 hours, the rectifier shall be operated at this ambient temperature and shall be capable of meeting the performance requirements of this specification.

4.6.2 *High temperature test.*—The rectifier shall be placed in a heat chamber, at an ambient temperature of +160° F, for a period of at least four hours. The temperature shall then be decreased to +125° F, and the rectifier allowed to attain thermal equilibrium. After exposure to the +125° F, ambient temperature for at least 24 hours, the rectifier shall be placed in operation, at this ambient temperature, and shall be capable of meeting the performance requirements of this specification. The operating temperatures of the plates in the selenium stacks shall be recorded at half-hour intervals and shall not exceed 37° F above ambient at any time interval.

4.7 **Operational test.**—Rectifier RA–63–( ) shall be operated from a 105–125 volt, 50 to 60 cycle, single-phase a-c power source and shall charge 6 fully discharged lead-acid cells in series (2 batteries BB–55 in series) starting at an initial rate of 15 to 16 amperes. The operating temperature of the selenium rectifier discs shall be recorded at half-hour intervals during the operation of the rectifier, until the

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batteries are brought to a fully charged condition as indicated by a specific gravity reading of 1.280. This test shall be conducted between 68° F and 86° F. The operating temperature of the selenium discs shall not exceed 37° F above ambient temperature at any time interval. The rectifier shall meet performance requirements without overheating of any of the parts.

4.8 **Load test.**—Rectifier RA–63–( ) shall be subjected to the following test: The rectifier shall be operated from a 115-volt, 50 to 60 cycle, single-phase a-c power source. With a 0.6-ohm resistance connected across the output, the d-c current shall read 15 amperes ±1 ampere with the control switch in the "HI-CHG" position. With a 2.1-ohm resistance connected across the output, the d-c current shall read 5 amperes ±0.5 ampere with the control switch in the "TR-CHG" position.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *For domestic shipment.*—Rectifier RA–63–( ) shall be preserved and packaged in accordance with Method III of Specification MIL–P–116.

5.1.2 *For export shipment.*—Preserve and package Rectifier RA–63–( ) in accordance with procedures specified for the designated methods as prescribed in Specification MIL–P–116, and further described in the following paragraphs.

5.1.2.1 *Technical manual.*—Package each technical manual Methods IC–3, as follows: Inclose each technical manual within a close-fitting Type II Class b bag, conforming to Specification JAN–P–117. Properly seal closure.

5.1.2.2 *Rectifier RA–63–( ).*—Package each item Method IIb, as follows: Close case and secure fastening. Cushion each item on all surfaces with cells or pads or both fabricated of corrugated fiberboard, conforming to Specification 19–10 and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned item together with the required amount of desiccant as prescribed in Specification MIL–P–116, within a close-fitting suitable style and grade corrugated fiberboard box. Seal entire closure with gummed Kraft tape conforming to Specification UU–T–111. Blunt all corners of the box. Place the boxed item within a water-vapor-proof barrier, qualified for Class 1 application in accordance with the requirements of Specification MIL–B–131. Heat seal closure. Place the water-vapor-proofed item within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN–P–108. Secure the technical manual packaged as specified in 5.1.2.1 to the packaged item. Seal entire box closure, corners and joints with water-resistant pressure-sensitive tape, conforming to Specification JAN–P–127.

5.2 *Packing.*

5.2.1 *For domestic shipment.*—Rectifier RA–63–( ) shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, or corrugated fiberboard boxes conforming to Specification NN–B–591, NN–B–601, NN–B–621, and LL–B–631, respectively. Closures shall be made in accordance with the applicable box specification.

5.2.2. *For export shipment.*—Place a quantity of the items packaged as specified in 5.1.2.2 within a wood cleated plywood or nailed wood box conforming to the requirements of Specifications JAN–P–105 and JAN–P–106 respectively. Fabricate the boxes to fit contents snugly. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification *only* for direct shipment to ports.

5.3 *Marking for shipment.*—Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL–STD–129. The nomenclature shall be Charger, Battery, Metallic (Rectifier RA–63–( )).

6. NOTES

6.1 Intended use.—Rectifier RA–63–( ) is intended for use as a compact, portable, selenium plate, full-wave bridge rectifier to convert 105–125 volt, 50 to 60 cycle, single-phase

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a-c power to direct current power suitable for charging 6 lead-acid storage cells in series.

6.2 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether domestic or overseas shipment is required.
(c) Samples required. (See 3.1.)
(d) Marking and shipping of samples.
(e) Acceptable quality level (see 4.1.1 (b), 4.1.2, and Table I).
(f) Technical literature required.

6.3 **Nomenclature.**—The parenthesis in the nomenclature will be deleted or replaced by a letter, identifying the particular design, for example, Rectifier RA–63–W. As soon as possible after award of contract, the contractor should apply to the contracting officer for such information (see 1.1).

6.4 **Contractual requirements.**—It is recommended that the following be included in the contract:

6.4.1 *Additional testing.*—It should be understood that additional inspection and nondestructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials, parts, subassemblies, or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.4.2 *Waiver of preproduction samples.*—When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; production of the item under the new order shall be concurrent with, or follow without interruption, production on the current order; and current production is in strict compliance with specified requirements for the new equipment, or describing any deviation authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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